Methodology on the Multi-Page Visually Rich Document Understanding Frontier: A Literature Review

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Degree: BACS (Honours)

Abstract

Multi-page visually rich document understanding (MP-VRDU) has matured along four architectural families: page-to-document encoders, backbone-centric multimodal LLMs, retriever-generator pipelines, and agentic pipelines. On top of these, an increasingly elaborate inventory of training-free orchestration techniques drives the current frontier. This review synthesises the literature on both, with disproportionate weight on the retriever-generator and agentic strands. A consistent pattern recurs across recent advances: progress has come from localised methodological refinements at the retrieval, reasoning, orchestration, and training-side stages of an otherwise stable architectural substrate, rather than from new architectures or datasets. Building on this observation, the review consolidates the open problems on the agentic frontier and motivates a methodology research direction targeting one or more existing systems, evaluated head-to-head against their published numbers on shared benchmarks.

1 Introduction

Documents are the principal medium through which organisations, governments, and the scientific community record and exchange information. Contracts, financial reports, scientific papers, technical manuals, regulatory filings, and presentation decks are pervasive and structurally heterogeneous, interleaving dense text with tables, figures, diagrams, and complex layouts [1; 2]. Manually extracting information from such documents is slow, expensive, and error-prone, motivating a sustained research effort to build automated document visual question answering systems that can answer natural-language questions over rendered document pages.

For single-page inputs, this problem is largely solved. Document-specialised systems such as Qwen2.5-VL [3] and mPLUG-DocOwl1.5 [4] approach or exceed human performance on DocVQA [5], InfographicVQA [6], and ChartQA [7] datasets. Recent frontier multimodal LLMs such as Claude Opus 4.7 [8] and Gemini 3.1 Pro [9] can now read a rendered page directly from pixels, treating the document as an image and matching or exceeding human reading on single-page and even short multi-page documents provided the input resolution is sufficient. Real-world documents, however, almost always extend beyond a single page, and the evidence required for a realistic question is typically scattered across several pages or distributed across heterogeneous elements [1; 10; 2]. Three structural challenges follow from the move to multi-page inputs. First, encoding each page already consumes hundreds to thousands of visual tokens at the resolutions required to read fine text [11], so multi-page input rapidly exhausts the context window of even long-context large language models (LLMs) and multimodal LLMs (MLLMs); quadratic-cost self-attention [12] compounds the compute burden further. Second, answering a question often requires resolving cross-page semantic and logical dependencies, including section hierarchy, multi-page tables, and figure-to-text references. Third, explainability becomes considerably harder, as a model must not only produce a correct answer but also surface which pages or passages support it.

Recent surveys cover visually rich document understanding (VRDU) primarily at the single-page level, organising methods by OCR dependency [13; 14]. The multi-page frontier, however, has developed its own architectural taxonomy and an increasingly elaborate inventory of inference-time techniques that are inadequately captured by an OCR-versus-no-OCR axis. This review focuses exclusively on multi-page VRDU and gives proportional weight to the families that now dominate the literature: retriever-generator pipelines and agentic pipelines. Section 2 first reviews the single-page substrate that multi-page systems inherit. Sections 3–7 then survey the multi-page literature across architecture, multimodal representation, training, training-free strategies, and datasets, and the final two sections (§§8–9) consolidate the resulting open problems and motivate a methodology research direction aimed at improving existing retriever-generator and agentic systems on the benchmarks they share.

2 Background: Single-Page VRDU

Multi-page VRDU systems inherit most of their per-page mechanisms from the single-page literature, recently surveyed by Ding et al. [13]. The dominant axis at the single-page level distinguishes two architectural families. **OCR-dependent frameworks** preprocess the page with off-the-shelf OCR or PDF parsers, producing recognised text and bounding-box layout that is either fed into the LLM prompt directly or projected through an auxiliary multimodal encoder such as LayoutLMv3 [15], with text-layout fusion supplied through 2D positional encodings, layout-aware attention, or rule-based verbalisation of bounding boxes. **OCR-free frameworks** discard the parser and instead have a vision encoder read text directly from rendered page pixels, with the LLM trained to recover text and layout through pretraining objectives such as text recognition, grounding, and captioning [16; 17; 18]. The two families trade different costs: OCR-dependent systems inherit cumulative parser errors but avoid expensive visual pretraining, whereas OCR-free systems handle handwriting and complex layout end-to-end but require high-resolution input and large-scale pretraining.

Across both families, three orthogonal design decisions recur and persist into the multi-page setting. Text is recovered as OCR strings in the prompt, through OCR-augmented multimodal encoders, or as a training objective for the vision encoder itself. Layout is encoded as 2D positional features, as verbalised bounding boxes in the prompt, or as supervised pretraining tasks such as text grounding. Visual tokens are managed through low-resolution patch embeddings, high-resolution shape-adaptive cropping [18], or compression via resamplers, Q-Formers, and convolutional reducers [19; 20]. Single-page systems make these choices assuming the entire document fits in a single backbone input; the multi-page setting reformulates each into a question of which page or region to attend to next.

3 Architectural Landscape

The OCR axis from single-page VRDU (§2) does not resolve the design dimensions introduced by multi-page documents. Inputs frequently exceed the backbone's context window, and the evidence required for a single answer is sparsely distributed across pages and modalities, forcing architectural choices about *how* and *where* cross-page evidence is selected and integrated, largely independently of OCR usage. Through this evidence-management lens, the multi-page literature divides into four recurring families: page-to-document encoding models, backbone-centric MLLM adaptation, retriever-generator pipelines, and agentic pipelines. The four are presented below in roughly the order they entered the field.

3.1 Page-to-Document Encoding Models

Page-to-document encoding models process each page through a shared layout-aware encoder such as LayoutLMv3 [14] or DocFormerv2 [21] that fuses OCR text, bounding-box layout, and visual features within the page, then connect per-page representations through cross-page attention [10; 2], per-page summary tokens [1], or recurrent state propagation [22]. By preserving page boundaries and cross-page reading order in the architecture itself, this design provides a strong inductive bias for documents with regular structural templates such as multi-page forms and reports. Every page must nonetheless be jointly encoded in a single forward pass, and the page-to-document compression step discards fine-grained evidence as inputs grow [10; 22]. Subsequent families address these pressures by replacing the dedicated cross-page encoder with more flexible mechanisms.

3.2 Backbone-Centric MLLM Adaptation

Backbone-centric MLLM adaptation removes the dedicated cross-page encoder entirely, instead processing the document inside a pretrained multimodal LLM whose context window absorbs cross-page information directly. Adaptation strategies range from training only lightweight projectors [23] or visual-token compression modules [11], to fine-tuning the backbone itself through continued pretraining [24], instruction tuning, or reinforcement learning [25]. This design avoids external retrieval or orchestration components and benefits directly from the MLLM's instruction-following and reasoning capabilities. However, the backbone's context length now caps total document size, fine-grained signals depend on the fidelity of token compression [26], and document structure such as section hierarchy and figure-caption linkage must be recovered implicitly from pixels [27]. These limits motivate moving evidence selection out of the backbone altogether.

3.3 Retriever-Generator Pipelines

Retriever-generator pipelines decouple evidence selection from answer generation: a retrieval module returns a compact subset of textual chunks, pages, or graph nodes, and a generator produces an answer conditioned on those units [28; 29]. Because only the selected subset enters the generator, the family lifts the context-length and token-compression limits of backbone-centric models and supports documents that exceed any single backbone's context window. The retrieval and generation stages may be implemented by single modules or distributed across multiple specialised components [30; 31; 32], but the overall control flow remains a fixed pipeline executed once per query, which exposes the retrieved evidence as a partial audit trail. The design assumes that relevant evidence is rankable by query similarity in a single retrieval step, so evidence missed at retrieval is unrecoverable at generation, and questions that require aggregating multiple weakly relevant pages frequently fail [33]; performance is further bounded by upstream component quality, since chunking, OCR, and modality mismatches propagate directly into the answer. These pressures motivate replacing the fixed pipeline with adaptive, multi-step reasoning.

3.4 Agentic Pipelines

Agentic pipelines replace the fixed retrieval pipeline with iterative evidence acquisition: one or more LLM controllers issue successive tool calls for retrieval, navigation, cropping, or OCR, and each subsequent decision is conditioned on the outcomes of earlier observations [34; 35]. Roles such as inspection, synthesis, and adjudication may be distributed across heterogeneous backbones [36; 37], but the defining property is that the trajectory itself adapts to question difficulty rather than following a fixed control flow; multi-component systems with predetermined execution order, even when described as "agents", remain within the retriever-generator family. By lifting the single-retrieval-step bottleneck of retriever-generator systems, this design allows compute to scale with question difficulty rather than document length and exposes the trajectory as an inspectable audit trail. However, the variable-length action sequence makes inference cost difficult to bound before execution, controllers remain sensitive to prompt phrasing and may commit early to wrong evidence with silent failure, reproducibility is complicated by stochastic generation, and the system inherits the recall ceiling of every retriever, parser, or OCR module it calls.

4 Multi-Page Multimodal Representation

While §3 categorises systems by where cross-page evidence is processed, the design challenges of MP-VRDU also differ substantially across modalities. Text often exceeds the backbone's context window and is sparsely distributed across pages; visual encoding cost grows with both page count and per-page resolution, making full-document high-fidelity encoding impractical; and cross-page document structure such as section hierarchies, multi-page tables, and cross-references is not captured by the per-token layout signals inherited from single-page VRDU. The three subsections below examine these modalities in turn, focusing on what distinguishes their handling in MP-VRDU from the single-page substrate reviewed in §2. Modality fusion is a cross-cutting concern across all three, but SP and MP systems differ less here than in the modality-specific challenges above, so it is treated as out of scope for this review.

4.1 Text Modality

The pressures specific to multi-page text are twofold: documents often exceed the backbone's context window, and even when they fit, the answer occupies a small fraction of a long input where attention is diluted and key statements are easily lost in the middle [38]. Responses cluster into three patterns. **Per-page aggregation** encodes each page in detail and combines the results into compact cross-page summaries through attention or memory tokens [1; 10; 2]. **Whole-document compression** keeps every page within a single backbone context by aggressively reducing per-page text tokens through resamplers, layout-aware compressors, or token-reduction modules [11; 39]. **Retrieve-then-read** abandons full-document encoding entirely, indexing text at chunk or page level and passing only a retrieved subset to the reader; agentic variants extend this into multiple selection-reading rounds [40; 41; 35; 32]. Each pattern fails in a characteristic way: per-page aggregation discards token-level detail at the summary step, whole-document compression sacrifices fine-grained reading on individual pages, and retrieve-then-read makes evidence missed at retrieval unrecoverable at the reading stage. The third failure mode is the most consequential for the present project, since it is shared by every retriever-generator and agentic system surveyed in §6.

4.2 Visual Modality

Multi-page visual encoding must remain fine-grained enough to recover charts, tables, and handwritten content that OCR cannot, yet scalable enough to apply across documents spanning tens or hundreds of pages. This tension is absent in the single-page setting, where one page can simply be encoded at the resolution required. The literature manages it through three patterns. **Aggressive token compression** keeps every page available in context but reduces per-page visual tokens through resamplers or convolutional reducers [11; 39], preserving coverage at the cost of per-page detail. **Selective high-resolution reading** restricts expensive encoding to a small subset of pages surfaced by retrieval or controller-driven inspection, sending those high-fidelity tokens into the reader while leaving other pages unread [34; 42]. **Retrieval-only visual embeddings** decouple encoding from reading altogether: page images are encoded purely to produce retrieval embeddings, and detailed reading of selected pages is handed to a separate text or vision module [28; 29]. The corresponding failure modes parallel those of the text modality: compression loses fine elements, selective reading inherits the recall ceiling of the selector, and retrieval-only embeddings rank layout-similar pages without confirming that the answer-bearing region is present. They reinforce the same theme that every method here treats the input document as the universe of visual evidence.

4.3 Layout and Cross-Page Structure

The structural problem distinctive to multi-page VRDU is *cross-page* structure: section hierarchies that span multiple pages, information referenced or continued across page boundaries, and document-level relationships such as figure-to-text or table-to-caption linkages that no longer fit within a single page's view. Within-page mechanisms inherited from single-page VRDU, such as 2D positional encodings, bounding-box coordinates, and reading-order signals attached to individual tokens [13], describe only per-page geometry, and per-token boxes do not encode cross-page relationships. End-to-end pretraining of cross-page structure has been attempted [11; 24] but is limited by the cost of supervision at this scale. The dominant route is therefore to expose parsed structural information to the model directly, through three patterns. **Structure-aware chunking** uses a parser to extract section trees and heading hierarchies, then segments documents along semantic boundaries so retrieved units respect section scope rather than sliding-window cuts [41]. **Hierarchical retrieval indices** route queries first to coarse units such as sections or chapters and then to fine-grained chunks within the selected branch [43; 44]. **Graph-based traversal** converts headings, entities, and cross-references into a knowledge graph, allowing controllers to follow typed edges rather than relying on dense similarity alone [30]. All three depend on parser quality, so errors in heading detection, table-boundary segmentation, or reading-order extraction propagate directly into every downstream retrieval and navigation step. This dependence is a structural fragility the project inherits, but, as §8 argues, it is not the limit this project sets out to address.

5 Training Strategies

How multi-page systems are trained tracks closely with the architectural family they sit in, and the literature has converged on a small set of strategies. This review treats them compactly: the trajectory of interest for this project is the move from monolithic supervised pretraining toward retrieval- and reasoning-targeted objectives, which sets up the training-free orchestration that follows in §6.

5.1 Parameter-Efficient Tuning on Frozen Backbones.

The most common strategy reuses strong pretrained LLMs or large vision-language models (LVLMs) and updates only a small number of task-specific components such as LoRA [45], projectors, adapters, layout tokenizers, or task-specific heads, leaving the bulk of pretrained parameters frozen. LayTokenLLM [46] keeps the LLM backbone frozen and introduces a lightweight layout-tokenisation module together with LoRA-based adaptation; CREAM [40] extends a similar idea to retrieval-augmented settings; and MoLoRAG [33] fine-tunes only the retrieval engine over a page graph.

5.2 Curriculum and Multi-Stage Training.

Direct adaptation of single-page models to multi-page objectives tends to underperform on cross-page dependencies, so many works adopt progressive curricula that start from simpler local tasks. DocSLM [27] moves from image-OCR alignment through image- and document-level compression to instruction tuning; mPLUG-DocOwl2 [11] follows a related OCR-free staged strategy, first ensuring that compressed visual tokens preserve text and layout, then pretraining on multi-page parsing.

5.3 Self-Supervised Retrieval Training.

Because retrieval-augmented pipelines are bounded by retriever quality, a growing line of work trains dedicated retrieval components directly. VDocRAG [29] introduces self-supervised representation-compression objectives that align dense visual document representations with OCR-derived textual content, and DFVC [47] trains only a lightweight cross-page fusion adapter on top of a frozen VisRAG or ColQwen backbone. This subline matters here because the visual retrievers it produces are the same components that an external-knowledge agentic system would deploy against a non-document corpus.

5.4 Reinforcement Learning for Reasoning.

Most recent and most directly relevant to the agentic frontier, several systems use reinforcement learning to optimise multi-step reasoning. DocR1 [25] introduces Evidence Page-Guided GRPO, rewarding the model for first identifying evidence pages and then generating an answer; Doc-V* [34] combines supervised trajectory distillation with GRPO over a composite reward; and MACT [36] extends RL to multi-agent settings with mixed per-agent and global rewards. These methods demonstrate that reasoning-quality signals, and not only answer correctness, can be optimised end-to-end, which becomes a prerequisite for training agents whose tools include both intra-document and external retrieval.

6 Training-Free Strategies

Training-free orchestration is the locus of the current MP-VRDU frontier and the substrate on which this project builds, so it warrants the most detailed treatment of any section in this review. These strategies improve performance without updating backbone parameters, ranging from offline document-side preparation (parsing into a graph, tree, or outline) to inference-time orchestration through prompting, tool use, retrieval refinement, and multi-agent coordination. They dominate the retriever-generator and agentic families introduced in §3, and the three stages surveyed below, retrieval and navigation, inference-time reasoning, and multi-agent decomposition, correspond directly to the methodological loci at which the project's eventual intervention will be inserted.

6.1 Retrieval and Navigation Techniques

Training-free retrieval uses off-the-shelf encoders, document structure, or LLMs as routers and re-rankers to select page or chunk evidence. Because retrieval quality bounds every downstream reasoning method, the choice of retrieval signal dominates system behaviour on long documents, and it is the most important locus at which an external evidence source could be introduced.

Sparse Hierarchy-Aware Search. The earliest training-free retrieval strategy in this corpus parses the document into an explicit textual structure such as section trees, heading hierarchies, or tree-formatted outlines, and retrieves through term- or structure-anchored matching. Retrieved chunks preserve their parent context, and section-level search and fetch tools allow controllers to navigate the parsed hierarchy directly [41; 35]. More recent variants extend the structure to dual hierarchies covering in-page chunks and cross-page topological summaries, addressing the fragmentation that single-tree designs leave unresolved [43]. The strength is structural awareness without learned components; the weakness is that sparse signals match on lexical or structural overlap and miss semantic connections and paraphrased queries.

Dense Embedding Search. Surface-form sensitivity motivated the shift to dense retrieval, where pages or chunks are encoded into learned representations and queries match on semantic similarity. Most MP-VRDU systems adopt vision-language late-interaction encoders such as ColPali [48], ColBERT [49], and ColQwen [50], which embed the query as a sequence of contextual token vectors and each page as a bag of patch embeddings, scored by MaxSim aggregation [51; 33; 32; 42; 37; 28].

Token-to-patch alignment is particularly important in this setting because a single question token may need to align with a figure caption, table cell, or paragraph buried in a long document. Adaptive selectors further scale the number of retrieved pages with the score distribution rather than using a fixed top-K [52]. Such methods handle paraphrase and cross-modal queries well but remain structure-blind: sections can be split across embedding chunks, and cross-page references are not represented as first-class objects. Critically for this project, these visual retrievers are well-developed and ready-to-use, which is what makes their application to external corpora a near-immediate engineering possibility.

Joint Multi-Signal Search. Several systems combine sparse and dense signals at the retrieval stage rather than committing to a single channel. Typical combinations include dense coarse retrieval filtered by iterative LLM-based re-ranking, page embeddings re-ranked against LLM-generated summaries under persistent working memory, and parallel text and image retrieval tracks whose candidates are aggregated by a downstream agent [40; 42; 32]. These combinations improve recall but still score each candidate passage in isolation, so retrieval cannot follow logical links, cross-references, or multi-hop dependencies that span passages.

Graph-Based Traversal. Graph approaches address the cross-passage gap by indexing the document as a graph and navigating it through edges. Variants differ in how nodes and edges are defined: knowledge graphs over passages and document-structure nodes traversed by an instruction-tuned LLM [30]; page graphs in which VLM-assigned logical relevance is combined with semantic similarity for multi-hop traversal [33]; and per-query graphs whose chunk nodes are anchored to LVLM-generated query nodes so the structure adapts to the question rather than being fixed at index time [44]. Graph methods reach references that pure scoring cannot, but incur construction overhead, edge-quality dependence, and traversal cost that scales with hop count. All four retrieval patterns share the same closed-corpus assumption identified in §3: every index, embedding, and edge is constructed from the input document itself.

6.2 Inference-Time Reasoning Strategies

On top of retrieval, the literature reasons about retrieved evidence through prompting, sampling, and action-selection mechanisms applied at inference. The contrast between the two patterns described below, a fixed evidence buffer versus an adaptive tool loop, is the contrast on which the project's agentic direction depends.

Chain-of-Thought and Self-Improvement. Chain-of-thought (CoT) prompting [53] elicits intermediate reasoning steps that integrate retrieved text, tables, and figures across pages before producing a final answer. Reasoning runs over a fixed evidence buffer assembled by a prior retrieval step, so the reasoner cannot acquire new evidence once decoding begins. Extensions add structured prompts over hierarchical retrieval output [43], self-refinement and self-verification loops that emit refined queries when evidence is judged insufficient [42], and document-level self-consistency with uncertainty-based selection [27]; trained variants internalise the CoT trajectory through reinforcement learning [25]. The shared limit is the closed evidence buffer: questions whose evidence sits outside the retrieved set remain unreachable regardless of how the trace is sampled or verified.

ReAct and Tool-Augmented Reasoning. ReAct [54] replaces the closed buffer with an action loop, alternating reasoning steps with tool calls whose observations feed the next step. In multi-page VRDU the tools retrieve, fetch, or inspect document pages, letting the reasoner gather evidence during inference. Variants instantiate the loop with InfoNCE-guided sub-query refinement [51], outline-driven section-level search and fetch [35], intent-conditioned action selection trained on trajectories [55], and thumbnail-level document maps with working-memory summaries [34]. ReAct is the dominant agentic pattern in the corpus and the natural point at which an external-retrieval action would be inserted: every implementation cited above already exposes an arbitrary tool interface, and the loop itself is agnostic to whether a given tool reads inside or outside the document. The shared costs are tool latency that grows with rounds, prompt sensitivity at each step, and fixed-resolution reading of any fetched page.

6.3 Multi-Agent Decomposition

A complementary strand decomposes the work across multiple specialised LLM and VLM agents, such as modality readers, retrievers, judges, and synthesisers, coordinated through a pipeline, controller, or critic loop. Three sub-patterns recur, ordered roughly by increasing flexibility.

Modality-Specialised Pipelines. The simplest configuration runs dedicated text and image agents in parallel after retrieval, with a synthesiser fusing their outputs [32], or splits work between a text filter, a visual extractor, and a modal fuser that applies confidence-conditional rules across modalities [31]. Parallel pipelines improve modality coverage compared to single-track retrieval, but lack an error-correction stage, so an incorrect output from one branch can dominate aggregation when the other branch lacks counter-evidence.

Verification and Adjudication. A second sub-pattern inserts an explicit verifier between proposal and answer. Adjudication-based variants sample multiple candidate answers and route them through an adjudicator that selects the most consistent conclusion [56]; actor-reviewer variants pair the answering agent with a reviewer that cross-checks through complementary tools and writes disagreements into a shared reflection memory [35]. Adjudication catches errors that fusion cannot, but the verifier typically draws on the same backbone family as the actor, so shared blind spots persist.

Adaptive Orchestration. The most flexible designs treat the loop itself as adaptive, allocating compute per question and per role. ViDoRAG [37] cycles a seeker, an inspector, and an answerer with a Gaussian-mixture model setting per-query K bounds from the similarity distribution; MACT [36] decomposes the loop into planning, execution, judgement, and answer agents with agent-wise adaptive test-time scaling, separating the judge from the corrector to avoid the same agent excusing its own errors. Adaptive orchestration improves the accuracy-latency trade-off but pays in latency that scales with both rounds and agents. Despite the rapid proliferation of multi-agent designs, the verification and critique mechanisms used inside MP-VRDU agents remain comparatively shallow next to those developed in the broader LLM-agent literature, an asymmetry that §8 and §9 take up as a candidate locus for methodological improvement.

7 Datasets and Evaluation

Dataset and benchmark choices both shape and constrain what a methodology project can demonstrate in MP-VRDU, which is why this review treats them as a single concern rather than as a neutral inventory.

Pretraining data reuses general-purpose document archives such as IIT-CDIP [57] and DocStruct4M [4] originally designed for single-page tasks; genuinely multi-page corpora have only recently emerged through MP-DocStruct1M, Doc-750K, and PaperPDF [11; 24; 58]. Fine-tuning has shifted to bespoke synthetic generation per model, with strong LVLMs writing question-answer or trajectory data over scraped PDFs as per-page annotation does not scale to long documents [23; 42; 25; 55]. Benchmarks have moved from mid-length extractive evaluation on MP-DocVQA, DUDE, and SlideVQA [1; 59; 60] to long-context multimodal multi-hop evaluation on MMLongBench-Doc, LongDocURL, and M-LongDoc [61; 62; 63], with retrieval and page-localisation increasingly scored alongside answer accuracy [64; 25; 56]. MP-DocVQA, MMLongBench-Doc, DUDE, DocVQA, and InfographicVQA dominate cross-model comparison.

Two structural features of this landscape matter for the project's framing. First, a small number of benchmarks (MP-DocVQA, MMLongBench-Doc, LongDocURL, DUDE) appear in roughly half or more of the surveyed papers and constitute the de-facto axes for cross-model comparison, which means a methodology contribution can be measured head-to-head against a substantial competitor set provided it reports on this same surface. Second, heavy reuse of the same benchmarks across training mixes and evaluation suites [61; 62] raises a contamination concern that any methodology project must address explicitly through ablations and multi-benchmark reporting. Both features are revisited in §9 when specifying the target systems and evaluation surface.

8 Open Problems and the Agentic Frontier

The training-free machinery surveyed in §6 is increasingly elaborate, yet headline accuracy on the field's long-document benchmarks [61; 62] remains well below human performance, and the strongest published systems cluster near a shared ceiling rather than separating sharply. The problems below account for much of that remaining gap and are ordered by how directly improvement on each would register on shared benchmarks.

8.1 Recent Progress in Agentic Pipelines

Agentic systems have produced the clearest qualitative gains over the past two years. By replacing the single-pass retrieve-then-generate control flow with iterative evidence acquisition, they raise the recall ceiling that bounds non-iterative RAG, expose the trajectory itself as an inspectable audit trail of which pages, regions, and modalities were consulted, and allow compute to scale with question difficulty rather than document length. They have also been the first family in which explicit verification and adjudication, allocated across distinct agent roles, are used to catch errors that single-branch fusion cannot. Importantly for a methodology-research direction, each of these gains was achieved through a localised intervention at one or two stages of an otherwise standard agentic pipeline, indicating that the same stages remain open to further refinement.

8.2 Outstanding Problems

Several open problems nonetheless persist across the surveyed corpus, and most are sharpened rather than resolved by the move to agentic control. The first two below are the most actionable as targets for methodology research, since each names a discrete weakness in an identifiable set of systems. The last two are diagnostic concerns that constrain how any improvement can be measured and reported.

Retrieval as the Binding Constraint. Retrieval performance now defines the upper bound of answer accuracy across the retriever-generator and agentic families [38]. Broadening the retrieved set introduces noise and attention dilution, while narrowing it makes evidence missed at retrieval unrecoverable. Existing approaches suffer from brittle chunking, OCR and parser error propagation, modality mismatch between retriever and generator, and per-passage scoring that ignores cross-reference and logical links [33; 37]. Iterative retrieval [33; 42] partially raises this ceiling but inherits the recall of every retriever and parser it calls. Several recent systems attribute their largest remaining error sources to retrieval [33; 34; 42], making retrieval-stage interventions the most direct route to headline-number improvement on the field's shared benchmarks.

Reasoning over Selected Evidence. Once evidence is retrieved, the reasoning stage in most surveyed systems is a single chain-of-thought pass [53] over the selected buffer, occasionally extended with a one-shot self-verification step [42]. Several reasoning-quality interventions developed in the broader LLM literature, including self-consistency sampling [65], tree-of-thoughts decoding [66], and verbal self-refinement [67; 68], have shown consistent gains on multi-hop and long-context reasoning but have only been partially or informally transferred to MP-VRDU [56; 35]. The reasoning stage is therefore a second locus where transplanting a well-studied technique could yield measurable gains at lower deployment cost than retrieval-side interventions.

Evaluation Integrity. The shift toward LLM-synthesised training data has compromised evaluation integrity at a structural level. MP-DocVQA appears in roughly 15 of 36 surveyed papers' fine-tuning mixes and 18 of their evaluation suites, and DUDE and DocVQA show similar overlap. Of the papers that introduce a synthesised training corpus, a substantial fraction source their documents from existing evaluation benchmarks, producing a circular loop in which a model is trained on a strong LVLM's behaviour over benchmark PDFs and later evaluated against the same PDFs under metrics frequently judged by the same LVLM [69]. This concern is diagnostic for the present project: it constrains the credibility of any headline-number improvement and motivates evaluation against multiple benchmarks, multi-seed variance reporting, and the use of model-disjoint or human judgements where possible.

Reproducibility of Stochastic Trajectories. Agentic pipelines produce stochastic trajectories whose tool calls and evidence selections vary across runs, complicating reproduction. Per-paper dataset curations and idiosyncratic metric choices further prevent direct system-to-system comparison. Any methodology contribution must therefore report multi-seed variance and target a benchmark on which the baseline numbers are themselves reproducible, which selects strongly for MMLongBench-Doc [61] and LongDocURL [62] as primary evaluation surfaces.

9 Targets and Loci for Methodological Improvement

The retrieval and reasoning stages of existing systems are the most directly actionable targets among the problems above, and the diagnostic constraints just listed narrow the credible evaluation surface to a small set of reproducible long-document benchmarks. This project pursues a methodology contribution at one of those stages, applied to a small set of leading agentic or retriever-generator systems and evaluated head-to-head against their published numbers on the benchmarks they share.

9.1 Candidate Target Systems

The choice of target system is constrained by three criteria: overlapping benchmark coverage with several published competitors, a reproducible methodology with publicly described components, and a sufficiently isolatable stage at which a novel technique can be inserted without retraining the backbone. Six agentic and retriever-generator systems published in 2025–2026 collectively satisfy these criteria and form the candidate set.

Agentic targets. ViDoRAG [37] composes a seeker, an inspector, and an answerer in an iterative loop with per-query K set by a Gaussian-mixture model over similarity scores. DocAgent [35] pairs an actor with a reviewer that cross-checks through complementary tools, writing disagreements into a shared reflection memory. Doc-V* [34] combines thumbnail-level navigation with high-resolution inspection of selected regions, supervised through RL with composite rewards over answer correctness, evidence recall, and format. MDocAgent [32] runs parallel text and image agents with a final synthesiser, providing a modality-specialised pipeline against which routing-side interventions can be evaluated. All four report on MMLongBench-Doc [61], and most additionally on LongDocURL [62] and either DUDE [59] or SlideVQA [60].

Retriever-generator targets. M3DocRAG [28] and VDocRAG [29] are the dominant single-pass visual-retrieval baselines. Both build on ColPali-style late-interaction retrievers [48] and report on MMLongBench-Doc as well as the open-domain M3DocVQA [70] and OpenDocVQA suites. SimpleDoc [42] couples ColPali retrieval with LLM-generated summary re-ranking under a persistent working memory, with published numbers on MMLongBench-Doc and LongDocURL that position it as a competitive non-agentic baseline. Because these three systems isolate retrieval and reading more cleanly than the agentic targets, they are the more attractive substrates for retrieval-stage interventions specifically.

The six systems collectively report on a small set of benchmarks dominated by MMLongBench-Doc, LongDocURL, MP-DocVQA, and DUDE, giving the project a clear head-to-head comparison surface. The intersection of their evaluations defines the leaderboard against which a methodological intervention can be measured.

9.2 Methodological Loci for Improvement

A novel methodology must intervene at a specific stage of the target pipeline. The literature reviewed in §6 segments cleanly into four such loci, each with a different cost profile and expected gain.

Retrieval and re-ranking. The retrieval stage admits the widest range of low-cost interventions: hybrid dense-sparse fusion, learned re-rankers, query rewriting and decomposition, iterative retrieval with confidence-based termination, and graph- or hierarchy-aware traversal [40; 42; 33; 43; 44]. Because several target systems attribute their largest remaining errors to retrieval (§8), an intervention at this stage has the most direct route to headline-number gains on shared benchmarks.

Reasoning over retrieved evidence. The reasoning stage in most target systems is a single chain-of-thought pass [53] with occasional self-verification. Techniques from adjacent literatures, including self-consistency sampling [65], tree-of-thoughts decoding [66], verbal self-refinement [67; 68], and step-wise process-reward verification [71], have not been systematically transferred to MP-VRDU. The expected gains here are typically smaller than at the retrieval stage but cheaper to deploy and easier to ablate, since the intervention can be applied without modifying the retrieval pipeline.

Orchestration and tool selection. The orchestration layer determines which tool an agentic system calls next. Existing target systems either use fixed action repertoires [35] or learned trajectory policies [34; 55]. Interventions at this layer include adaptive action selection conditioned on retrieval confidence, learned termination criteria, and tool-retrieval over a larger tool inventory. Gains are harder to predict in advance but cheaply ablated against the target systems' own action policies, and the verification literature [69; 68] supplies several drop-in critique mechanisms that the target systems do not currently use.

Training-side interventions. A novel training-side technique, such as retrieval-aware fine-tuning [72] or trained self-critique [73], is a higher-cost but potentially higher-payoff route. It requires synthesising or curating trajectories and re-training the relevant agent, both of which constrain the choice of target system to those with publicly released checkpoints. This locus is most appropriate when the chosen intervention requires behaviours that cannot be elicited through prompting alone.

9.3 Transferable Techniques from Adjacent Literatures

Each of the loci above has a small set of techniques developed elsewhere that the project may draw on. The text-only LLM literature has the most mature inventory and the cleanest transfer paths to MP-VRDU.

Self-consistency and tree-search decoding. Self-consistency [65] samples multiple reasoning chains and selects the majority answer, with consistent gains on multi-hop QA. Tree-of-thoughts decoding [66] generalises this to branching search with backtracking, giving stronger results on tasks that admit verifiable intermediate states. Neither has been systematically applied to MP-VRDU agents, though DocLens's sampling-adjudication [56] is a partial instance of the former.

Verbal self-refinement and reflection. Self-Refine [67] and Reflexion [68] use the same model to critique and rewrite its prior outputs, with measurable gains on reasoning and coding tasks. DocAgent [35] already implements a simpler reviewer-with-reflection loop, but the full Reflexion-style verbal RL has not been used to improve agentic MP-VRDU trajectories.

LLM-as-judge and process supervision. LLM-as-judge [69] and process-reward models [71] have become standard tools for fine-grained trajectory evaluation in text agents. In MP-VRDU they could either supply training signal for an RL fine-tune of an existing agent's policy or act as an inference-time verifier over candidate trajectories. Both uses are compatible with the target systems' existing interfaces.

Retrieval-aware training. Self-RAG [73] trains a single model to decide when to retrieve and to critique retrieved evidence, and RAFT [72] fine-tunes a generator to be robust to retrieval noise. Both methodologies are formulated for text-only RAG but transfer naturally to the visual-retrieval setting used by M3DocRAG, VDocRAG, and SimpleDoc, and would slot into the training stage of an existing target without changing its retrieval architecture.

9.4 Research Direction

The project commits to a methodology research direction with three constraints fixed and one substantive choice still open. It targets a subset of the candidate systems in §9.1, intervenes at one of the four methodological loci identified in §9.2, and reports headline-number comparisons against the targets' published values on at least MMLongBench-Doc [61] and LongDocURL [62], with multi-seed variance and ablations isolating the contribution of the proposed technique from the rest of the target pipeline.

The choice of specific technique remains open and will be made on the basis of empirical pilots: which of the techniques surveyed in §9.3, applied at which locus and on which target system, produces the largest reliable gain on the shared benchmarks. The contribution this review motivates is therefore narrow but specific in scope: not a new architecture, not a new dataset, but a measurable methodological improvement on an existing baseline, evaluated under conditions designed to surface the contribution cleanly and to satisfy the diagnostic constraints on evaluation integrity and reproducibility identified in §8.

10 Conclusion

Multi-page visually rich document understanding has consolidated around a clear architectural taxonomy and an increasingly elaborate body of training-free orchestration, with the retriever-generator and agentic families dominating the current frontier. The combined machinery of these families supplies adaptive evidence acquisition, modality-aware retrieval, and verification-based reasoning. The most fruitful recent advances have come from localised methodological refinements layered on top of a relatively stable architectural substrate [34; 37; 36], suggesting that further gains are reachable at the retrieval, reasoning, orchestration, and training-side loci consolidated in this review. The project this review motivates pursues that pattern: a methodology contribution at one of those loci, applied to a small set of leading agentic or retriever-generator targets and evaluated against their published numbers on shared benchmarks. The specific technique remains to be selected through empirical pilots, but the surrounding decisions of target system, methodological locus, and evaluation surface are committed and consistent with the trajectory of the field.

11 LLM Usage and Other Notes

LLM Usage. Large language models, specifically Claude Opus 4.7 and GPT 5.5, were used throughout the preparation of this review for spelling and grammar correction, sentence-level rewording, reading-flow refinement, and other writing-quality tasks. All conceptual content, argumentation, and selection of cited works are my own. The LLM was used as a writing aid rather than as a source of claims, and all factual statements, citations, and quantitative figures have been verified against primary sources before inclusion. Intellectual responsibility for the final document rests with the author.

Relationship to a Companion Survey Paper. Sections 3–?? were adapted from a survey paper on the MP-VRDU frontier of which I am the first author. The corresponding material was largely written by me prior to any co-author contributions to the survey. The adaptation in this review trims, re-tones, and reorients the survey material toward the methodology research direction motivated in §§8–9; the remaining sections (Introduction, Background, Open Problems, Targets and Loci, and Conclusion) are original to this document.

Supporting Materials. The following supporting artefacts were produced during the preparation of this review:

- Paper inventory and cross-reference spreadsheet (surveyed systems, dataset and benchmark coverage, evaluation metrics): `https://docs.google.com/spreadsheets/d/1ZoPeIvnZKkHOQN8G3Fjj9gY8jnLImZvcYQMJd3l6CajS78e/edit?usp=sharing`
- Project repository (literature review LaTeX source, build script, bibliography, and supporting code): `https://github.com/LeweiXu/MP-VRDU/`
- Overleaf project for the companion survey paper: *URL pending*.

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